Open-Motion User Manual

Engineering Mode

Product	Open-Motion blood-flow monitor — engineering / diagnostic features
Application version	1.4.0
Document revision	July 2026 — draft
Audience	Openwater engineering, manufacturing & support staff

DRAFT — not a controlled document. Engineering mode exposes diagnostics and maintenance actions that can alter device configuration and data output. It is intended for Openwater personnel only and must be disabled on systems handed to clinical or study users.

Contents

1. What engineering mode is

Engineering mode is a hidden, password-protected feature level that overlays extra diagnostics on either distribution (Clinical or Research) of the Open-Motion app: hardware maintenance controls, calibration and factory-test workflows, raw-data outputs, firmware update tooling, and plot/contact-quality diagnostics.

Everything documented in the Clinical and Research manuals continues to work unchanged; this manual covers only what engineering mode **adds**.

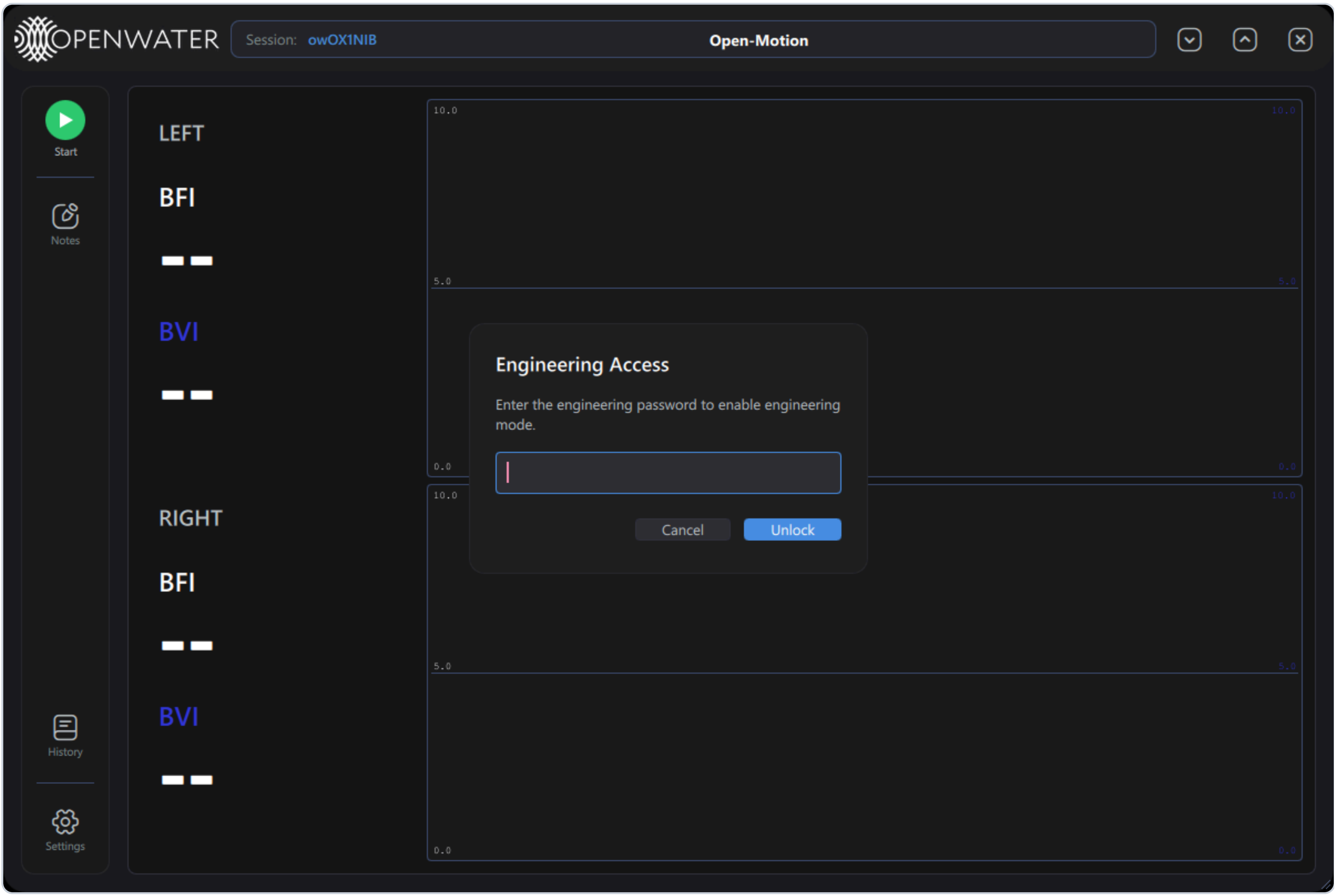
Feature map

Area	What engineering mode adds	§
Settings	**Engineering** card: console soft-reset, fans, debug logging, raw CSV, calibration & test	3
Settings	**Trace colors** row, **Beta Updates** switch	3.3, 6
Test / calibration	**Test Results** window with per-camera pass/fail table	4
Contact quality	**Force Dismiss** button, cause legend, split dot colors, detailed tooltips	5
Firmware	Firmware-update banner, per-device Update chips	6
Plots	Per-camera die temperature, **Profiler** HUD	7
Audit	Password-gated audit-log viewer	8
Scan Settings	Per-sensor fan toggles	9
Data output	Per-scan telemetry CSV; optional raw histogram CSVs	10

2. Enabling and disabling

Enable

1. **Double-click the Openwater logo** in the window header (no visible affordance — it is a hidden gesture).
2. The **Engineering Access** prompt opens:



3. Enter the engineering password (available from Openwater engineering — it is deliberately not printed in documentation) and press **Unlock** (or `Enter`). **Cancel**, `Esc`, or clicking outside closes without unlocking; a wrong entry shows "*Incorrect password*" and clears the field.

On success a toast confirms "*Engineering mode enabled.*" The setting **persists across restarts** (it is written to the local configuration overrides) and the unlock is recorded in the audit log as a `settings_changed` event.

Disable

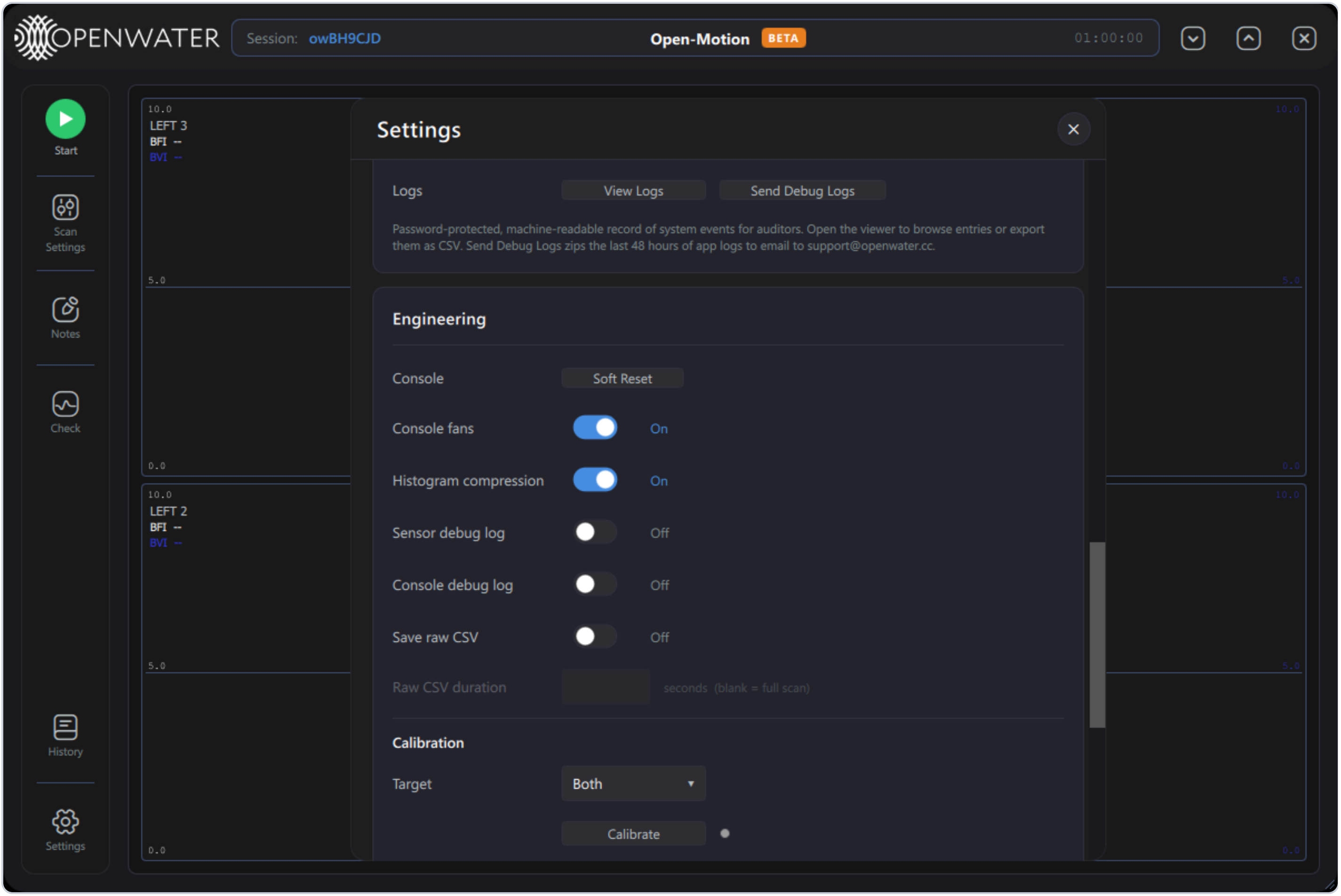
Settings → Engineering card → **Disable engineering mode** (toast: "*Engineering mode disabled.*"). Always disable before handing a system back to operators.

The same engineering password also gates three flows that are visible outside engineering mode: audit-log viewing (§8), scan deletion in History, and Calibrate (§3.2).

---

3. Settings → Engineering card

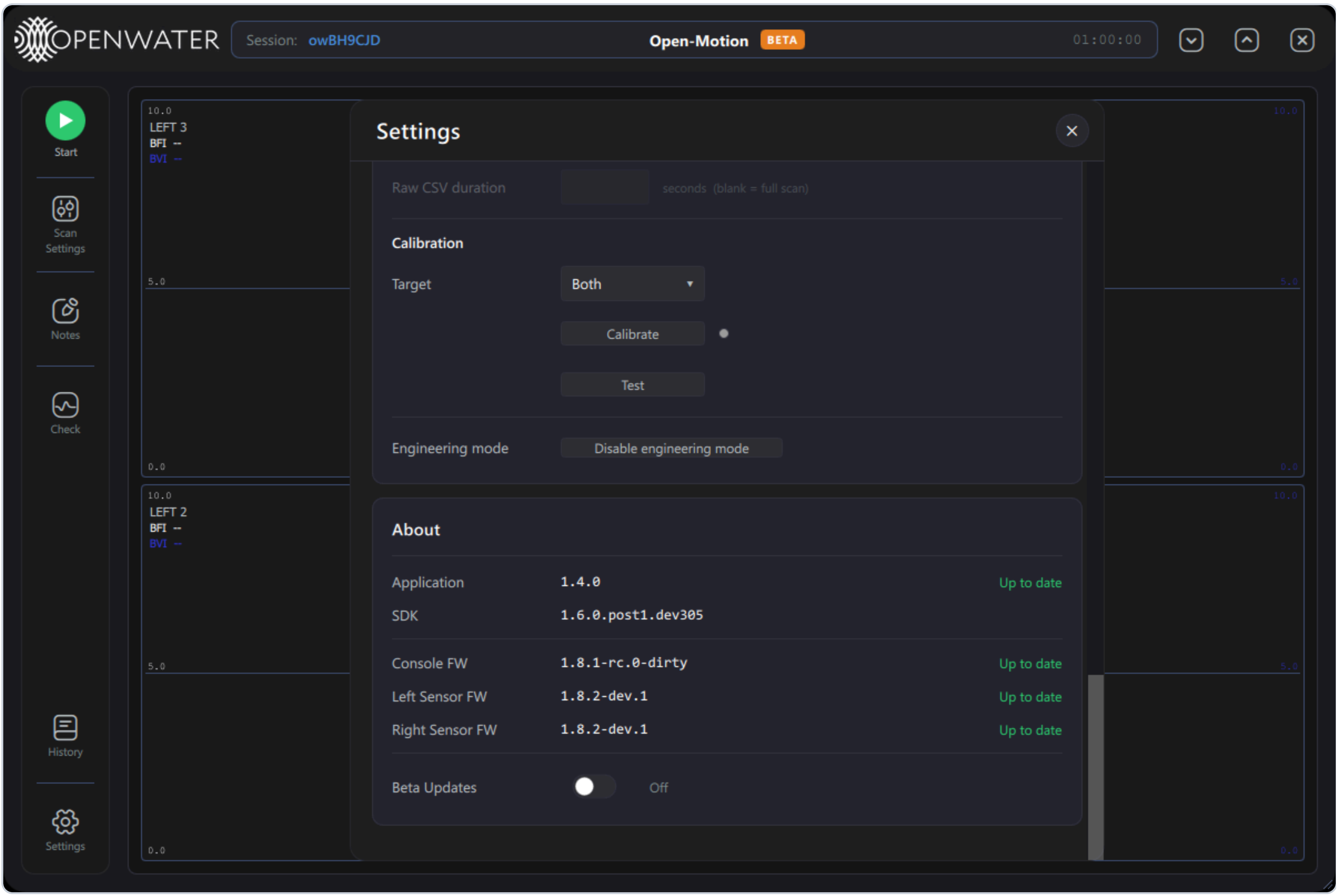
With engineering mode on, two new pieces appear in Settings: the **Engineering** card and extra rows elsewhere. The Engineering card:



Row	Control	What it does
Console	`Soft Reset` button	Reboots the console MCU (it drops off USB and reconnects a few seconds later).
Console fans	switch	Console cooling fans at full speed (On) or off. Only enabled while a console is connected.
Histogram compression	switch	Toggles the sensors' histogram-compression debug flag live (also persisted). Reduces USB bandwidth per frame; normally On.
Sensor debug log	switch	Streams sensor-firmware printf output into the app log (verbose; for diagnosing firmware issues).
Console debug log	switch	Same for the console firmware.
Save raw CSV	switch	Writes raw per-frame histogram CSVs during scans (see §10). Off by default — files are large.

Row	Control	What it does
Raw CSV duration	number field	Caps how many seconds of raw CSV are written per scan; blank = the whole scan. Enabled only when *Save raw CSV* is on.

3.1 Calibration & Test



Control	What it does
Target dropdown	Which side the workflow runs on: `Both`, `Left`, or `Right`.
Calibrate button	Runs the full calibration workflow (~15 s scan per side) and **writes the resulting calibration into the console EEPROM**. Password-protected. **Caution: this permanently alters the device's stored calibration** — run only per manufacturing/service procedure.
status dot + text	Next to Calibrate: blue = running, green = passed, red = failed, orange = aborted, with a status line ("Calibrating... (Ns / Ns)", "Calibration Passed/Failed/Aborted").
Test button	Runs the same measurement as a ~5 s **diagnostic only** — nothing is written to the device. Results appear in the Test Results window (§4).

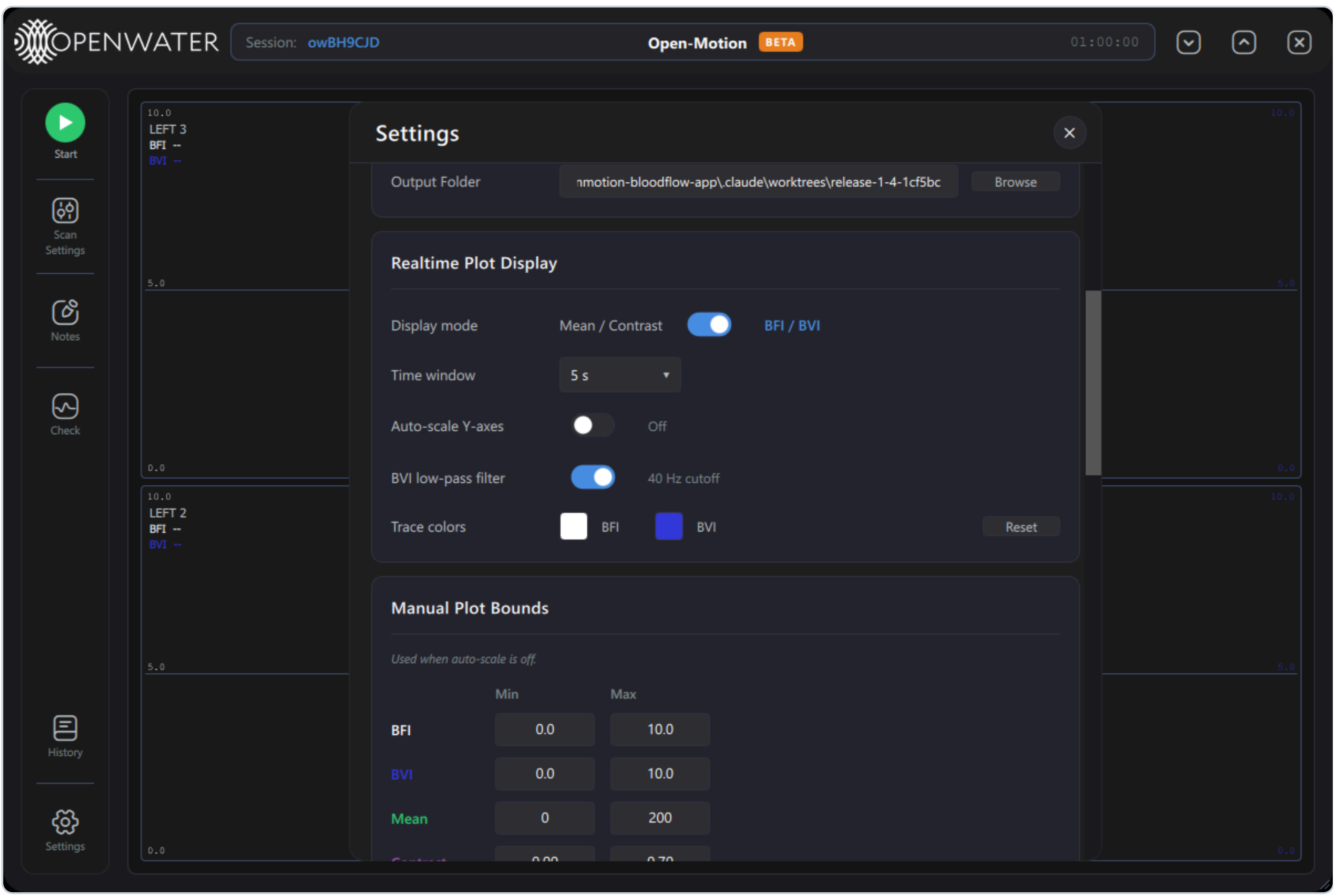
Control	What it does
Engineering mode	`Disable engineering mode` button (§2).

Both workflows briefly configure **all** cameras regardless of the scan pattern. While one runs, the app treats it like a scan for exit protection (closing the window warns first).

3.2 Password-gated actions elsewhere

- **History → Delete** — permanent scan deletion asks for the engineering password.
- **Settings → Audit Log → View Logs** — §8.

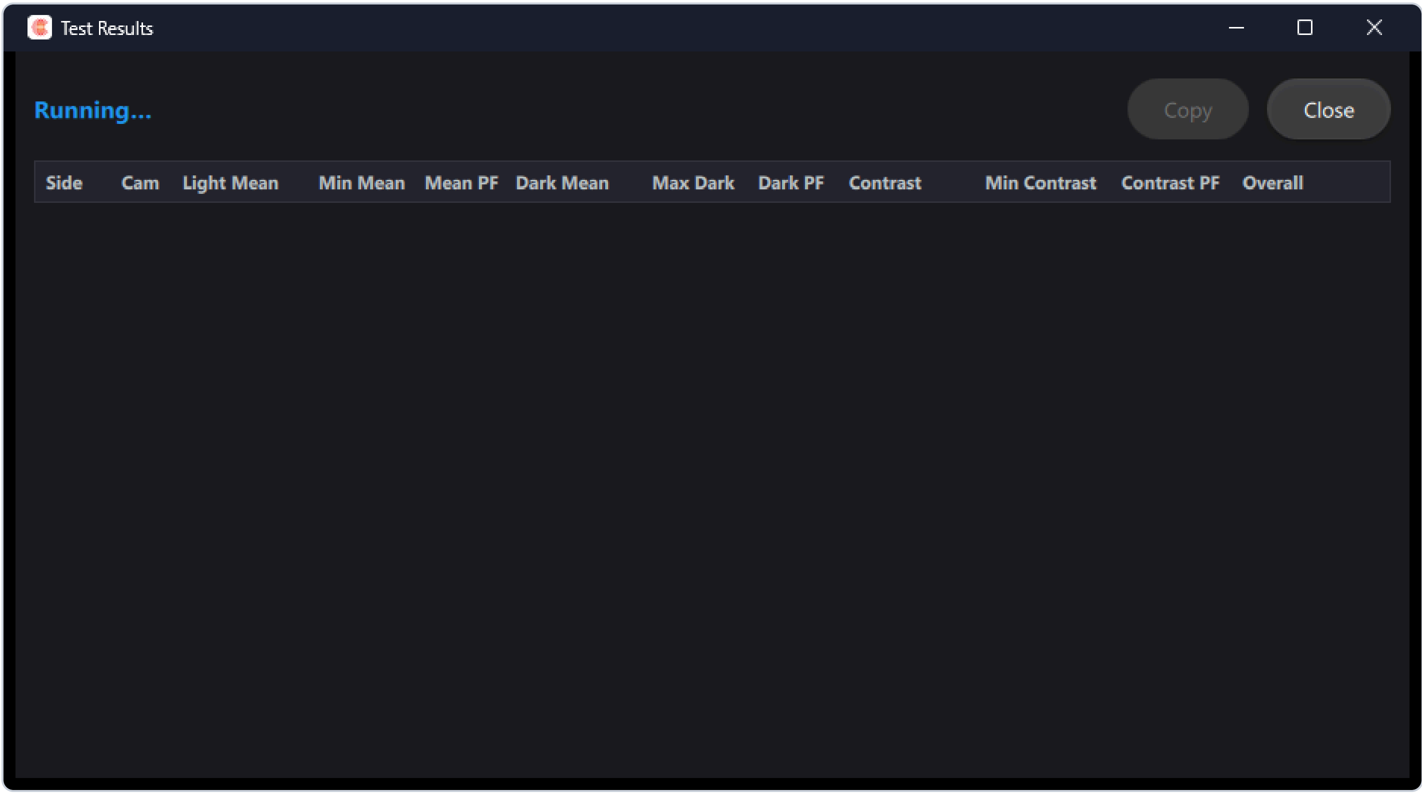
3.3 Trace colors (Realtime Plot Display)

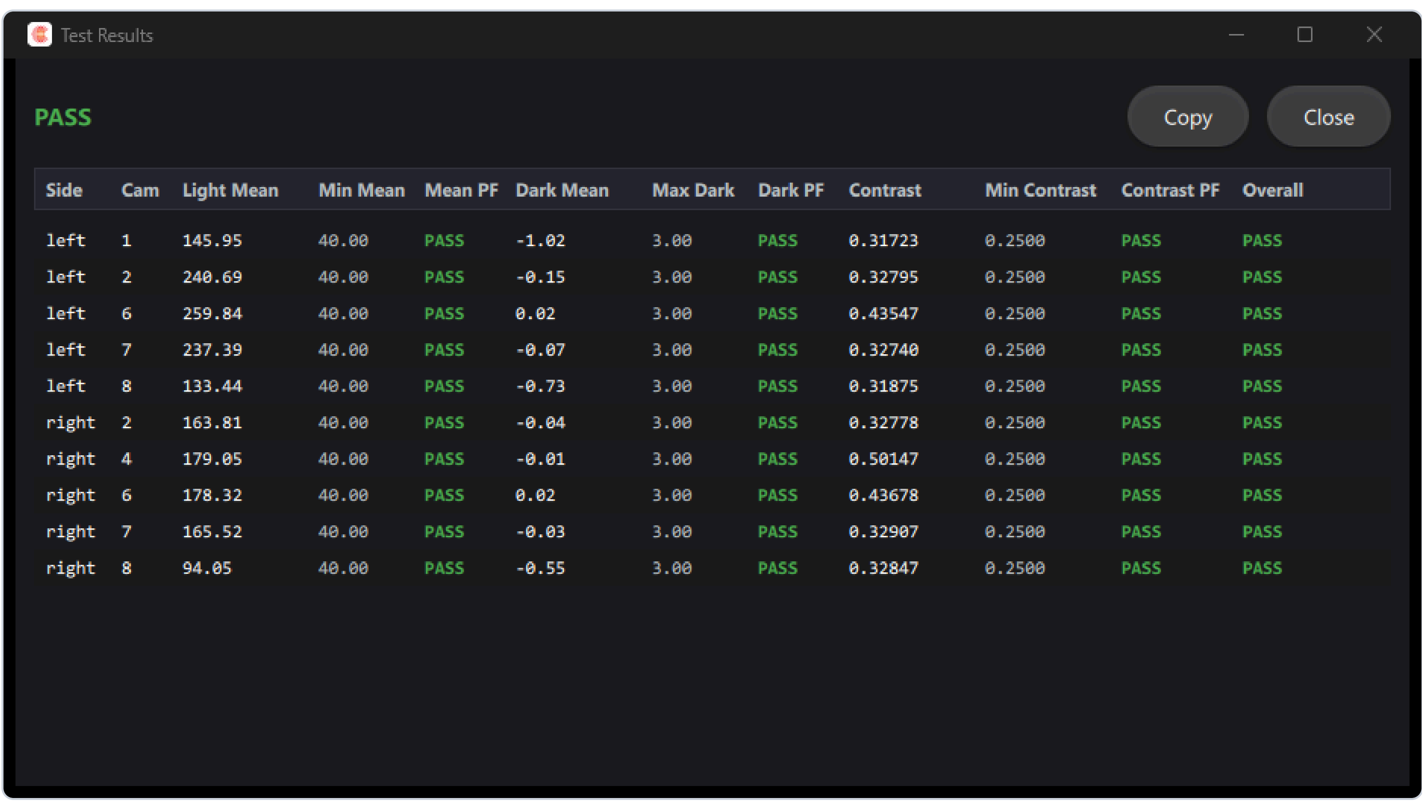


An extra row in *Realtime Plot Display*: click the **BFI** or **BVI** swatch to pick a custom trace color (system color picker), **Reset** restores the defaults. Colors are kept legible against the current theme automatically.

4. Test Results window

Launched automatically whenever a **Test** or **Calibrate** run starts (it is a separate window — it may sit on top of the main one):





Side	Cam	Light Mean	Min Mean	Mean PF	Dark Mean	Max Dark	Dark PF	Contrast	Min Contrast	Contrast PF	Overall
left	1	145.95	40.00	PASS	-1.02	3.00	PASS	0.31723	0.2500	PASS	PASS
left	2	240.69	40.00	PASS	-0.15	3.00	PASS	0.32795	0.2500	PASS	PASS
left	6	259.84	40.00	PASS	0.02	3.00	PASS	0.43547	0.2500	PASS	PASS
left	7	237.39	40.00	PASS	-0.07	3.00	PASS	0.32740	0.2500	PASS	PASS
left	8	133.44	40.00	PASS	-0.73	3.00	PASS	0.31875	0.2500	PASS	PASS
right	2	163.81	40.00	PASS	-0.04	3.00	PASS	0.32778	0.2500	PASS	PASS
right	4	179.05	40.00	PASS	-0.01	3.00	PASS	0.50147	0.2500	PASS	PASS
right	6	178.32	40.00	PASS	0.02	3.00	PASS	0.43678	0.2500	PASS	PASS
right	7	165.52	40.00	PASS	-0.03	3.00	PASS	0.32907	0.2500	PASS	PASS
right	8	94.05	40.00	PASS	-0.55	3.00	PASS	0.32847	0.2500	PASS	PASS

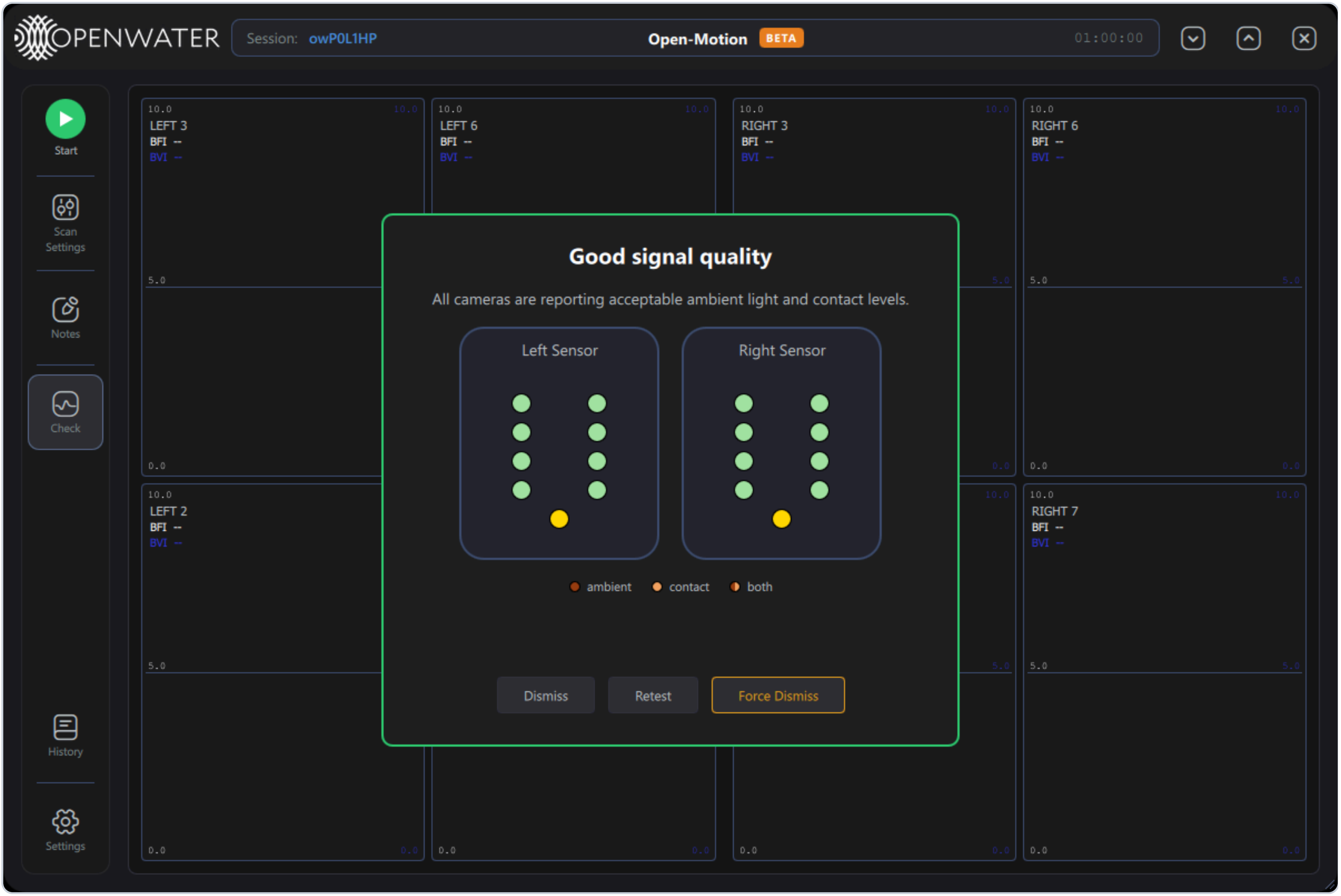
Element	Meaning
Header	`Running…` (blue) → `PASS` (green) / `FAIL — reason` (red) / `Aborted` (orange).
Copy	Copies the whole table as tab-separated text (for pasting into a report/sheet). Enabled once rows exist.
Close	Closes the window (it reopens on the next run).
Table	One row per tested camera: `Side, Cam, Light Mean, Min Mean, Mean PF, Dark Mean, Max Dark, Dark PF, Contrast, Min Contrast, Contrast PF, Overall`. Each `PF` cell is the pass/fail verdict of that metric against its configured threshold; `Overall` combines them.

Metrics: *Light Mean* = mean pixel level with the laser on (must exceed *Min Mean*); *Dark Mean* = residual level in laser-off frames (must stay below *Max Dark* — checks ambient-light leakage); *Contrast* = speckle contrast (must exceed *Min Contrast* — checks the optical path). Thresholds come from the application configuration (`ft_*` keys).

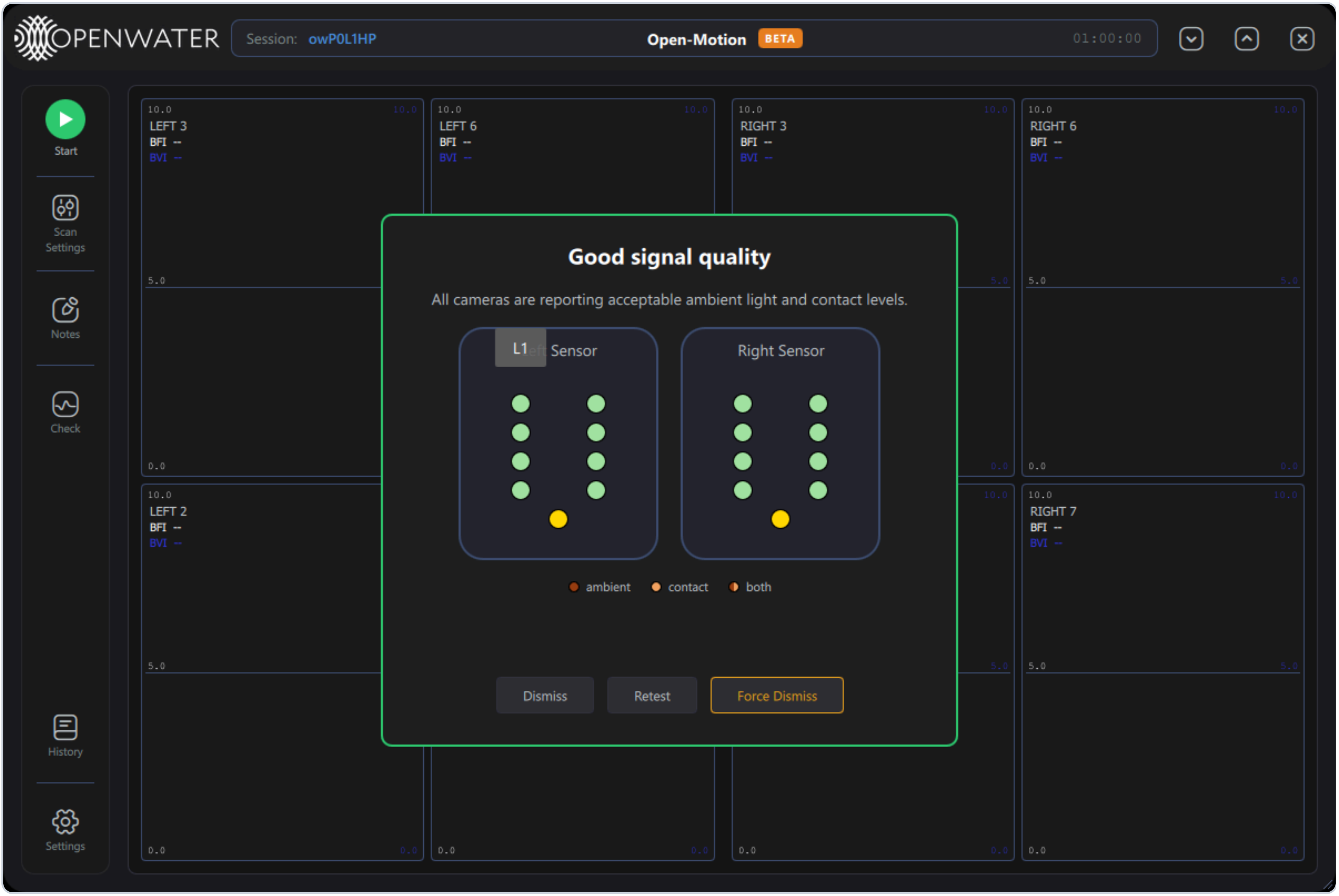
Each run also writes a CSV report and a JSON manifest under `data/calibrations/` (`test- ⟨timestamp⟩ .csv` / `.json`, `calibration- ⟨timestamp⟩ .csv`).

5. Contact-quality diagnostics

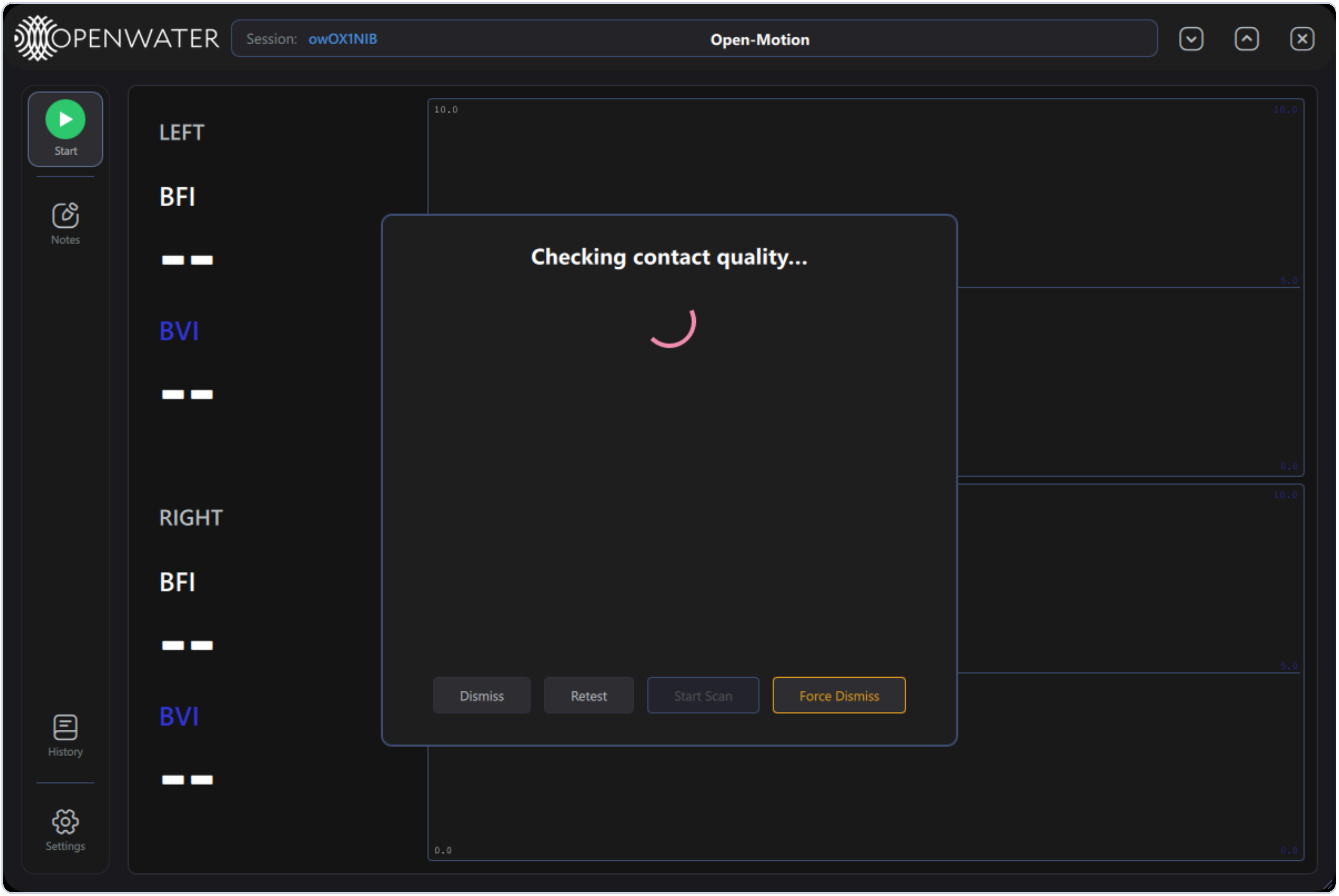
Engineering mode upgrades the contact-quality dialog (both the standalone Check and the clinical pre-scan gate):



- **Force Dismiss** (amber) — closes the dialog unconditionally, bypassing every contact-quality gate, **including while a check is still running** and during the clinical pre-scan flow. Diagnostic escape hatch only.
- **Cause legend** — `ambient` / `contact` / `both` swatches. Failing dots are colored by cause: dark orange = too much ambient light, sandy orange = poor skin contact; a dot split vertically shows both causes at once.
- **Tooltips with values** — hovering a dot shows the camera ID chip and, for failing cameras, the reason with the measured value in DN (digital numbers), e.g. "*Too much ambient light (12.4 DN)*":



- The footer row is visible even during the "checking" phase (so Force Dismiss is always reachable):



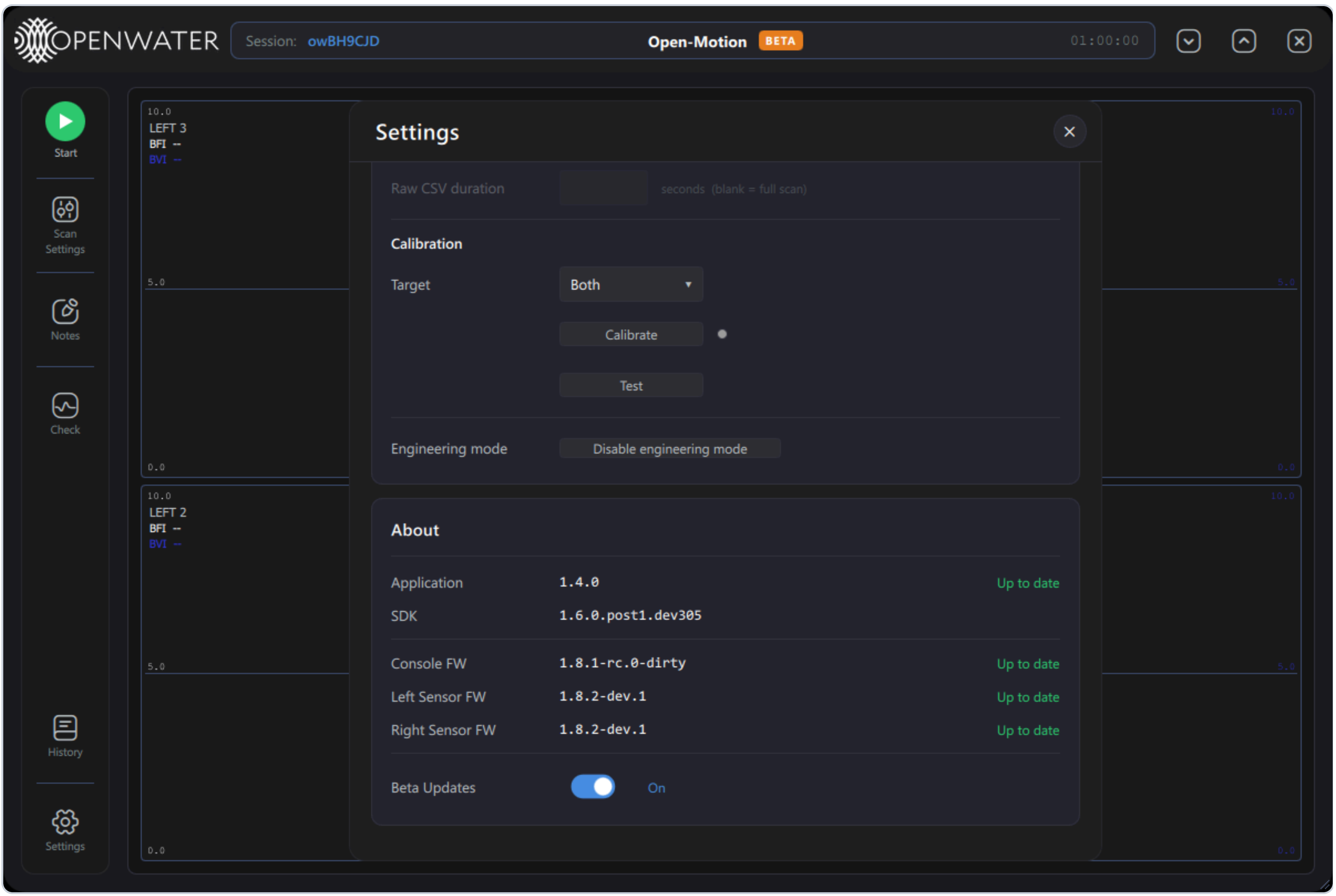
Clinical pre-scan check still running, with the engineering footer present — Start Scan stays disabled until the check finishes, but Force Dismiss works immediately.

---

6. Firmware & beta updates

Engineering mode owns device-firmware maintenance:

- **Firmware-update banner** — when any connected device's firmware is older than the latest release, a banner appears under the header: "*Device firmware update available*" with **View** (opens Settings at the firmware card) and ✕ (dismiss until the next detection).
- **About card chips** — each device row (Console FW, Left/Right Sensor FW) shows *Up to date* or an **Update** chip. Clicking a chip opens a **Confirm Firmware Update** dialog (Cancel / Update); during the update a progress line and bar track download → flash. Updates are refused while a scan is running.
- **Beta Updates** switch (About card, engineering-only):



On = update checks include dev/rc pre-releases instead of production releases only. The bench system above is already on the newest builds, so everything reads "Up to date".

After a firmware update, the device power-cycles/reconnects; wait for the About card to show the new version before scanning.

7. Plot diagnostics

7.1 Per-camera temperature

During a scan each plot cell gains an orange **die temperature** readout (top-right):

Typical steady-state camera temperatures run 50–75 °C depending on position.

7.2 Profiler HUD

The plot `···` menu gains a fourth switch, **Profiler**:

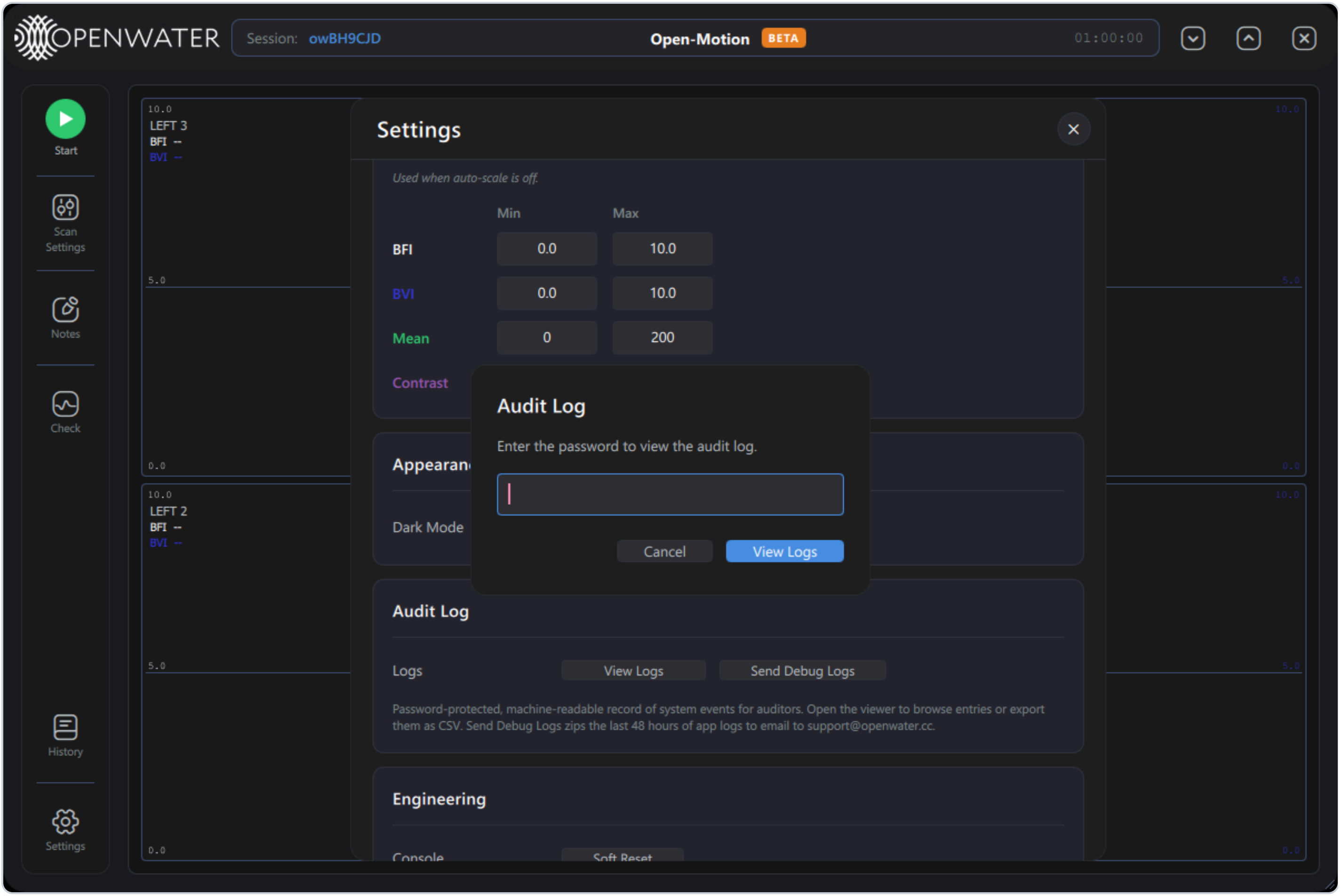
Turning it on shows a rendering-performance HUD in the bottom-left of the plot area:

Field	Meaning
`rate`	Incoming sample rate being plotted (Hz).
`tick`	Plot update interval (ms).
`canvas`	Average canvas repaint cost (ms).
`points`	Points currently drawn per repaint.

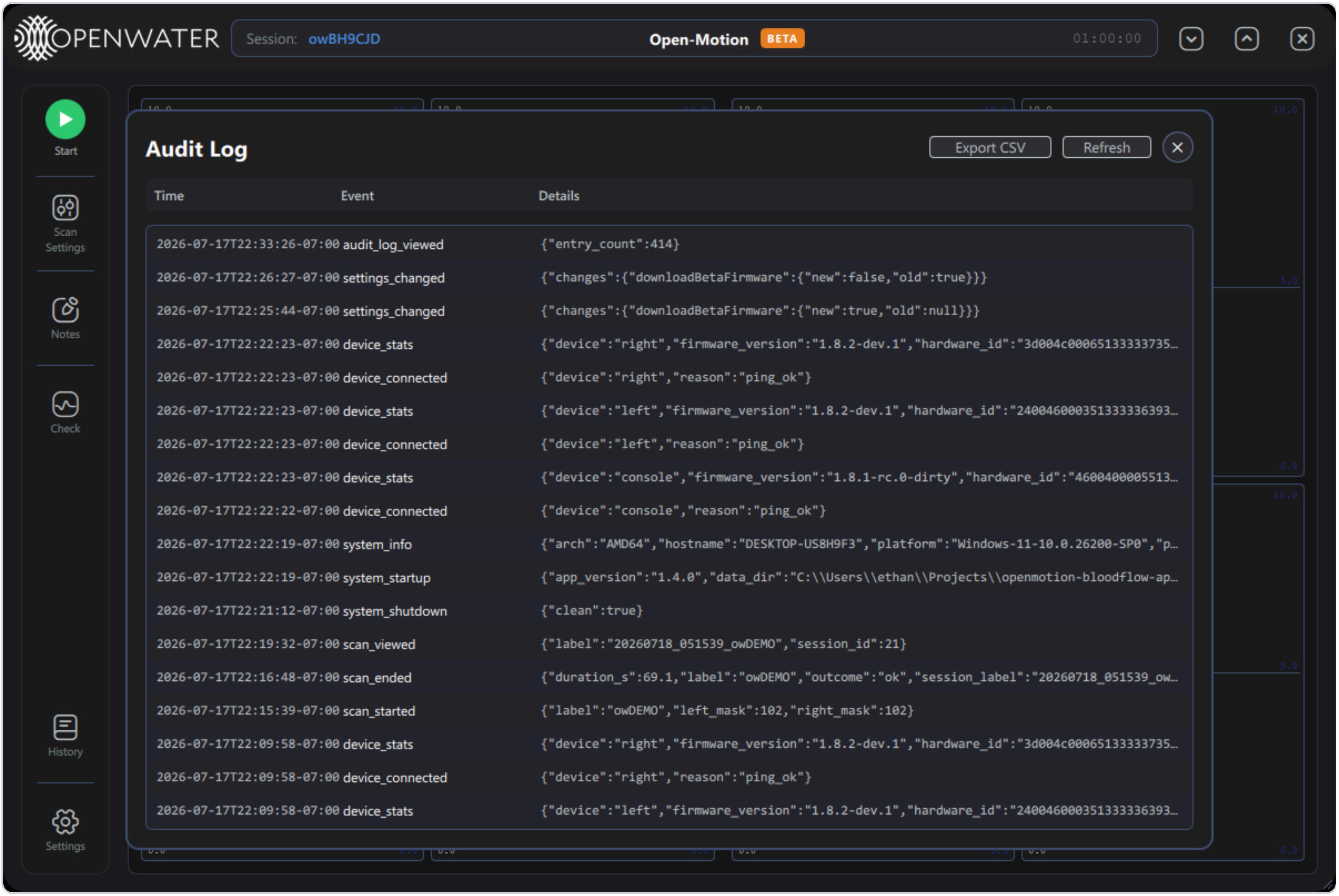
Use it to diagnose UI sluggishness on low-end hosts (e.g. with All-cameras patterns or long windows).

8. Audit-log viewer

Settings → Audit Log → **View Logs** asks for the engineering password:



and opens the viewer:



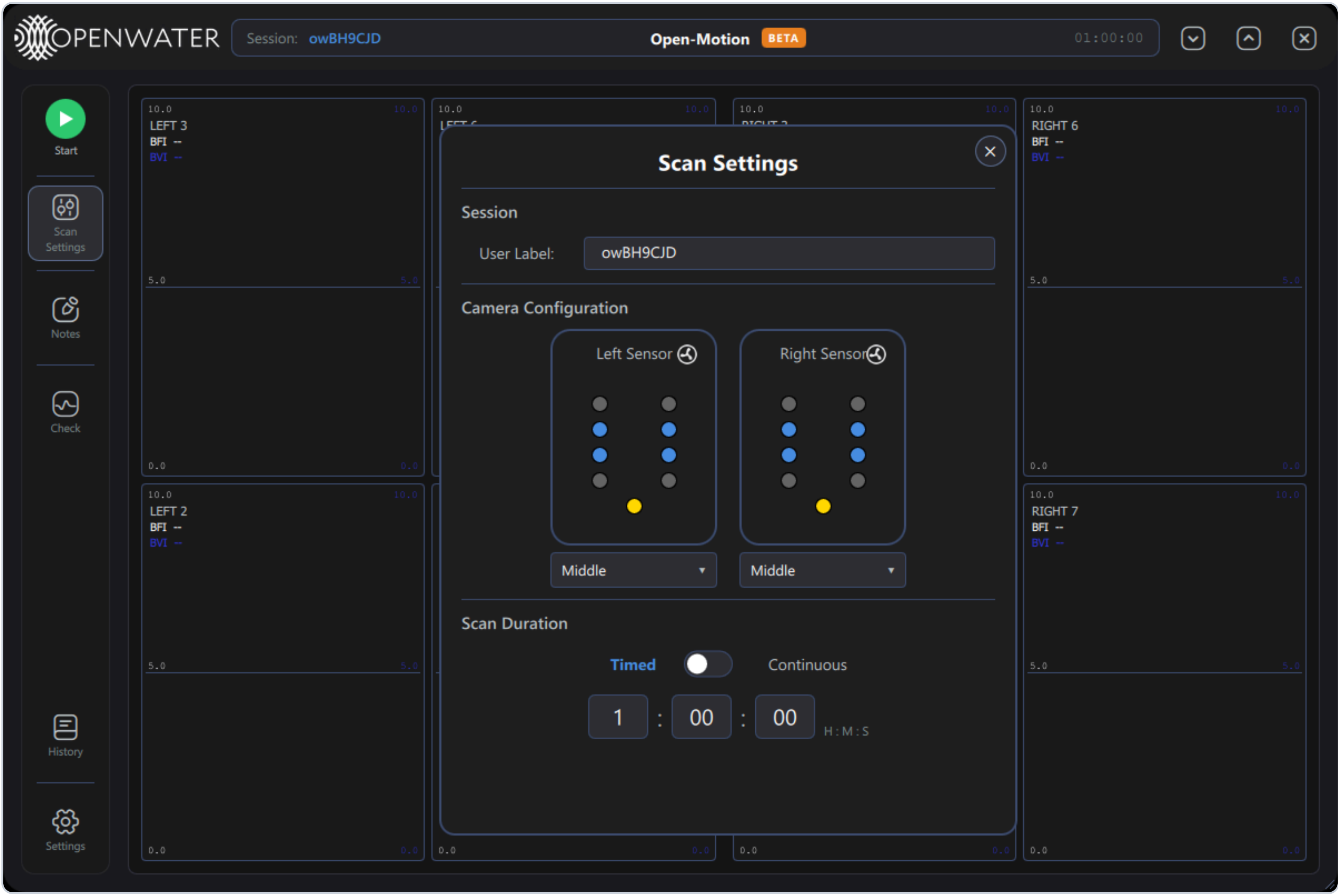
Control	What it does
Export CSV	Saves the audit log to a CSV file (save dialog, success toast).
Refresh	Reloads the newest entries (the viewer shows the latest 500).
×	Closes the viewer.

Recorded events include: `system_startup` / `system_shutdown` / `system_info`, `device_connected` / `device_disconnected` / `device_stats` (with firmware versions and hardware IDs), `settings_changed` (with old→new value diffs — including engineering-mode changes), `scan_started` / `scan_ended` / `scan_viewed`, `scan_deleted`, and `audit_log_viewed` (viewing the log is itself audited).

---

9. Sensor fan toggles

In **Scan Settings**, each sensor card gains a small fan icon next to its title:



Clicking it toggles that sensor module's cooling fan on/off (state is read back from the device when the dialog opens). The console's fans are controlled separately from the Engineering card (§3). Fans default to on; leave them on for normal operation — camera noise performance is temperature-dependent.

10. Engineering data outputs

With engineering mode enabled, scans can produce additional files next to the scan database (all under the Output Folder's `data/` directory):

Output	When it is written	File
Telemetry CSV	Every scan while engineering mode is on	`⟨scan_id⟩ _ ⟨label⟩ _telemetry.csv` — per-frame temperatures and system telemetry.
Raw histogram CSVs	Only when *Save raw CSV* is also on (§3)	`⟨scan_id⟩ _left/right_mask ⟨mask⟩ _raw.csv` — raw per-frame histograms; capped by *Raw CSV duration*. Large files.
Calibration / test reports	Each Calibrate/Test run	`data/calibrations/…` (§4).
Debug bundles	Send Debug Logs button	`data/debug-bundles/ ⟨timestamp⟩ .zip`.

Turning engineering mode off stops the telemetry and raw CSVs even if their switches are left on (clinical systems never produce them).

11. Configuration reference (advanced)

The app ships defaults in `config/app_config.json` and persists changes to a local overrides file (`app_config.local.json` in the writable data root). UI settings cover the common keys; a few engineering-relevant keys are config-file-only:

Key	Default	Meaning
`histoThrottle`	false	Drop histogram frames to reduce log spam during bench debugging.
`deferHistoSend`	true	Firmware-side deferral of histogram sends out of the frame ISR.
`commVerbose` / `verboseCommandHandling`	false	Log every UART/USB packet and MCU printf — extremely verbose.
`scanDataStallTimeoutSec`	3	Whole-scan watchdog: abort with E-303 if **no** camera delivers frames for this long (≤0 disables).
`cameraTempAlertThresholdC`	110	Camera over-temperature alert threshold.
`tecTripTempC`	40	Console over-temperature trip pushed to the console at connect.
`forceLaserFail`	false	**Debug only:** simulates a laser-safety trip; it latches a real interlock that blocks scanning until the console is power-cycled.
`writeCorrectedCsv`	false	Legacy corrected per-camera CSV per scan (redundant with the database).
`cq_*`, `ft_*`	—	Contact-quality and factory-test thresholds (per camera).

> **Never hand-edit `config/laser_params.json`.** It holds the laser-driver register baseline; incorrect values can produce wrong pulse widths or safety failures. It is locked baseline data owned by the laser/firmware teams.

12. Operational cautions

- **Calibrate writes to the console EEPROM** — run it only as part of a documented manufacturing/service procedure, with the correct target selected.
- **Force Dismiss** bypasses the contact-quality gate; data captured afterwards may be unusable. Never use it on subject scans.
- **Leave engineering mode disabled** on systems used by operators: it changes data output (telemetry CSVs), exposes destructive actions, and relaxes safety-adjacent gates.
- The engineering password is shared by the unlock, audit-log, delete and calibrate gates — treat it as controlled information.
